

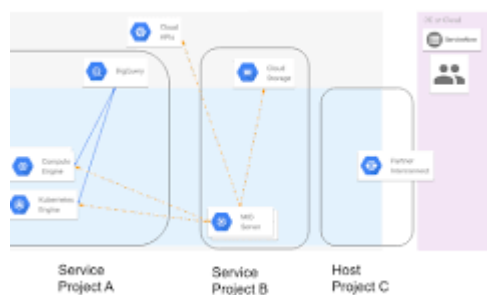
## Project Design Phase-II

### Technology Stack (Architecture & Stack)

|               |                                             |
|---------------|---------------------------------------------|
| Date          |                                             |
| Name          | Sridevi S                                   |
| Project Name  | Automated Network Request Management System |
| Maximum Marks | 4 Marks                                     |

#### Technical Architecture Overview

The proposed solution is built on the ServiceNow Now Platform as a cloud-based enterprise application. It connects user service requests with backend asset management through automated workflows, reducing manual effort and improving operational efficiency.



#### Structural Breakdown:

- **Frontend Tier (Local Client Interaction):** Users interact with a clean Service Portal built using modern web standards. Client-side processing handles UI dynamic logic, instant context retrieval, and validation filters.
- **Process Orchestration Engine (Cloud Logic):** Triggers server-side logic pathways via ServiceNow Flow Designer and Workflow Studio. This engine manages operational gates, rules processing, and line-manager approval loops.
- **Database Ledger Store (Persistence Layer):** Securely seals validated data inputs directly into custom enterprise database schemas without manual text copy-pasting.

**Table-1: Components & Technologies**

| S.No | Component           | Description                                                                                                           | Technology                                                                     |
|------|---------------------|-----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| 1.   | User Interface      | Customer-facing Web portal interface where users search, select request options, and view live stage progress badges. | ServiceNow Service Portal, HTML5, CSS3, JavaScript (ES6), Bootstrap Framework. |
| 2.   | Application Logic-1 | Client-side validation scripts, session properties query rules, and dynamic field visibility behaviors.               | ServiceNow Client Scripts, Catalog UI Policies, GlideUser (g_user) API.        |
| 3.   | Application Logic-2 | Server-side background execution workflows that intercept user checkout paths and check rule conditions.              | ServiceNow Flow Designer, Workflow Studio Actions, Business Rules.             |
| 4.   | Application Logic-3 | Cross-table routing calculations and relational approval mapping generation handlers.                                 | ServiceNow GlideRecord API, JavaScript Engine.                                 |
| 5.   | Database            | Local system data dictionary structures holding system transactional variables and workflow runtime metrics.          | ServiceNow Database Layer (built on MySQL/MariaDB foundation core).            |
| 6.   | Cloud Database      | Permanent, structural production relational table hosting the master infrastructure asset configurations.             | Custom System Schema Table ( <b>u_network_database</b> ).                      |
| 7.   | File Storage        | Secure storage for transactional audit history logs and physical office floor plan document attachments.              | ServiceNow Managed Cloud Attachment Storage (sys_attachment).                  |

| S.No | Component                       | Description                                                                                                          | Technology                                                                            |
|------|---------------------------------|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 8.   | External API-1                  | Identity contextual information query engine mapping incoming user sessions to corporate organizational directories. | ServiceNow internal GlideSystem session context lookup API.                           |
| 9.   | External API-2                  | Automated outbound notifications providing line managers with one-click decision-making functions.                   | ServiceNow Cloud Mail Server Engine (SMTP/IMAP Notification Engine).                  |
| 10.  | Machine Learning Model          | Optional optimization rule set used to automatically assign fulfiller tasks based on agent workload queues.          | ServiceNow Predictive Intelligence Engine (Optional Similarity/Clustering Framework). |
| 11.  | Infrastructure (Server / Cloud) | High-availability cloud host layer ensuring multi-tenant isolation, data separation, and secure runtime boundaries.  | ServiceNow Enterprise Cloud Platform-as-a-Service (PaaS) Instance.                    |

**Table-2: Application Characteristics**

| S.No | Characteristics          | Description                                                                                                                          | Technology                                                                        |
|------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| 1.   | Open-Source Frameworks   | Integration of universally supported front-end grid layouts to ensure responsive UI display across various client browser viewports. | Bootstrap CSS Framework, AngularJS (ServiceNow Core UI framework stack wrapper).  |
| 2.   | Security Implementations | Data encryption definitions safeguarding active sessions, access control parameters, role                                            | Advanced Encryption Standard (AES-256), Multi-Factor Authentication (MFA), Access |

| S.No | Characteristics       | Description                                                                                                                                     | Technology                                                                                                              |
|------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
|      |                       | profiles, and platform threat protection rules.                                                                                                 | Control Lists (ACLs), ServiceNow role-based access management.                                                          |
| 3.   | Scalable Architecture | Logical layer breakdown strategy that isolates user presentation elements from backend ledger transactions to manage fluctuating query traffic. | Multi-tier Cloud Platform-as-a-Service (PaaS) framework, microservices-inspired multi-node cloud application nodes.     |
| 4.   | Availability          | Resilient architectural planning guaranteeing zero service interruptions via continuous redundancy paths and fallback cluster instances.        | ServiceNow Advanced High Availability (AHA) Architecture, geo-redundant automated data center replication.              |
| 5.   | Performance           | Computational tuning to enforce minimal form load delays, sub-second execution speeds, and background workflow balancing processing.            | ServiceNow Client-Side Cache Optimization, Indexed Table fields on u_network_database, Asynchronous Background Workers. |